RESIDUAL GERBES MAP ISOMORPHICALLY WHEN THE FIELD-POINT SELF-PRODUCT DOES NOT CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3125-3140 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 3142-3167, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_stacks-properties.json (pb=1011, stmt 3125-3140, proof 3142-3167).

Editorial note. stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 04T4: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3167–3167. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-improve-unique-point: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2791–2802. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-residual-gerbe-functorial: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3101–3113. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-residual-gerbe-unique: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 3076–3086. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $x \in |\mathcal{X}|$ with image $y \in |\mathcal{Y}|$. Assume the residual gerbes $\mathcal{Z}_x \subset \mathcal{X}$ and $\mathcal{Z}_y \subset \mathcal{Y}$ of x and y exist and that there exists a morphism $\mathrm{Spec}(k) \to \mathcal{X}$ in the equivalence class of x such that*

$$\mathrm{Spec}(k) \times_{\mathcal{X}} \mathrm{Spec}(k) \longrightarrow \mathrm{Spec}(k) \times_{\mathcal{Y}} \mathrm{Spec}(k)$$

is an isomorphism. Then $\mathcal{Z}_x \to \mathcal{Z}_y$ is an isomorphism.

Proof. Let k'/k be an extension of fields. Then

$$\mathrm{Spec}(k') \times_{\mathcal{X}} \mathrm{Spec}(k') \longrightarrow \mathrm{Spec}(k') \times_{\mathcal{Y}} \mathrm{Spec}(k')$$

is the base change of the morphism in the lemma by the faithfully flat morphism $\mathrm{Spec}(k' \otimes k') \to \mathrm{Spec}(k \otimes k)$. Thus the property described in the lemma is independent of the choice of the morphism $\mathrm{Spec}(k) \to \mathcal{X}$ in the equivalence class of x. Thus we may assume that $\mathrm{Spec}(k) \to \mathcal{Z}_x$ is surjective, flat, and locally of finite presentation. In this situation we have

$$\mathcal{Z}_x = [\mathrm{Spec}(k)/R]$$

with $R = \mathrm{Spec}(k) \times_{\mathcal{X}} \mathrm{Spec}(k)$. See proof of Lemma lemma-improve-unique-point. Since also $R = \mathrm{Spec}(k) \times_{\mathcal{Y}} \mathrm{Spec}(k)$ we conclude that the morphism $\mathcal{Z}_x \to \mathcal{Z}_y$ of Lemma lemma-residual-gerbe-functorial is fully faithful by Algebraic Stacks, Lemma 04T4. We conclude for example by Lemma lemma-residual-gerbe-unique. □